

Exploratory Data Analysis and visualization for given data set

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from wordcloud import WordCloud
import seaborn as sns
```

```
df = pd.read_csv('location.csv')
```

```
print("First 5 rows of datasets: \n")
df.head()
```

↗ First 5 rows of datasets:

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	H
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	
2	2	2	Just finished an amazing workout! 🏃 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	
3	3	3	Excited about the upcoming weekend	Positive	2023-01-15	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	

Next steps:

[Generate code with df](#)

[View recommended plots](#)

[New interactive sheet](#)

```
print("\nData Info:\n")
df.info()
```

↗

Data Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 732 entries, 0 to 731
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Unnamed: 0.1    732 non-null   int64
1   Unnamed: 0      732 non-null   int64
2   Text            732 non-null   object
3   Sentiment       732 non-null   object
4   Timestamp       732 non-null   object
5   User            732 non-null   object
6   Platform        732 non-null   object
7   Hashtags        732 non-null   object
8   Retweets        732 non-null   float64
9   Likes           732 non-null   float64
10  Country         732 non-null   object
11  Year            732 non-null   int64
12  Month           732 non-null   int64
13  Day             732 non-null   int64
14  Hour            732 non-null   int64
dtypes: float64(2), int64(6), object(7)
memory usage: 85.9+ KB
None
```

```
print("Summary of Dataset: \n")
df.describe()
```

Summary of Dataset:

	Unnamed: 0.1	Unnamed: 0	Retweets	Likes	Year	Month	Day	Hour
count	732.000000	732.000000	732.000000	732.000000	732.000000	732.000000	732.000000	732.000000
mean	366.464481	369.740437	21.508197	42.901639	2020.471311	6.122951	15.497268	15.521858
std	211.513936	212.428936	7.061286	14.089848	2.802285	3.411763	8.474553	4.113414
min	0.000000	0.000000	5.000000	10.000000	2010.000000	1.000000	1.000000	0.000000
25%	183.750000	185.750000	17.750000	34.750000	2019.000000	3.000000	9.000000	13.000000
50%	366.500000	370.500000	22.000000	43.000000	2021.000000	6.000000	15.000000	16.000000
75%	549.250000	553.250000	25.000000	50.000000	2023.000000	9.000000	22.000000	19.000000
max	732.000000	736.000000	40.000000	80.000000	2023.000000	12.000000	31.000000	23.000000

```
print("Data Types of columns: \n")
print(df.dtypes)
```

Data Types of columns:

```

Unnamed: 0.1    int64
Unnamed: 0      int64
Text            object
Sentiment       object
Timestamp       object
User            object
Platform        object
Hashtags        object
Retweets        float64
Likes           float64
Country         object
Year            int64
Month           int64
Day             int64
Hour            int64
dtype: object
```

```
print("Missing values (if any): \n")
print(df.isnull().sum())
```

Missing values (if any):

```

Unnamed: 0.1    0
Unnamed: 0      0
Text            0
Sentiment       0
Timestamp       0
User            0
Platform        0
Hashtags        0
Retweets        0
Likes           0
Country         0
Year            0
Month           0
Day             0
Hour            0
dtype: int64
```

```
numeric_cols = df.select_dtypes(['int64', 'float64']).columns
corr_matrix = df[numeric_cols].corr()
```

```
print("Correlation matrix: \n")
print(corr_matrix)
```

Correlation matrix:

```

      Unnamed: 0.1  Unnamed: 0  Retweets    Likes    Year \
Unnamed: 0.1    1.000000    0.999995    0.388637    0.376208    0.101578
Unnamed: 0      0.999995    1.000000    0.388884    0.376472    0.100749
Retweets        0.388637    0.388884    1.000000    0.998482   -0.039982
Likes           0.376208    0.376472    0.998482    1.000000   -0.043415
Year            0.101578    0.100749   -0.039982   -0.043415    1.000000
Month           0.443013    0.443523    0.073265    0.066643   -0.314845
Day            -0.080101   -0.080480    0.009213    0.011489    0.021973
Hour            0.322371    0.322163    0.196955    0.195331   -0.087470

      Month    Day    Hour
Unnamed: 0.1  0.443013 -0.080101  0.322371
Unnamed: 0    0.443523 -0.080480  0.322163
Retweets      0.073265  0.009213  0.196955
Likes         0.066643  0.011489  0.195331
```

```

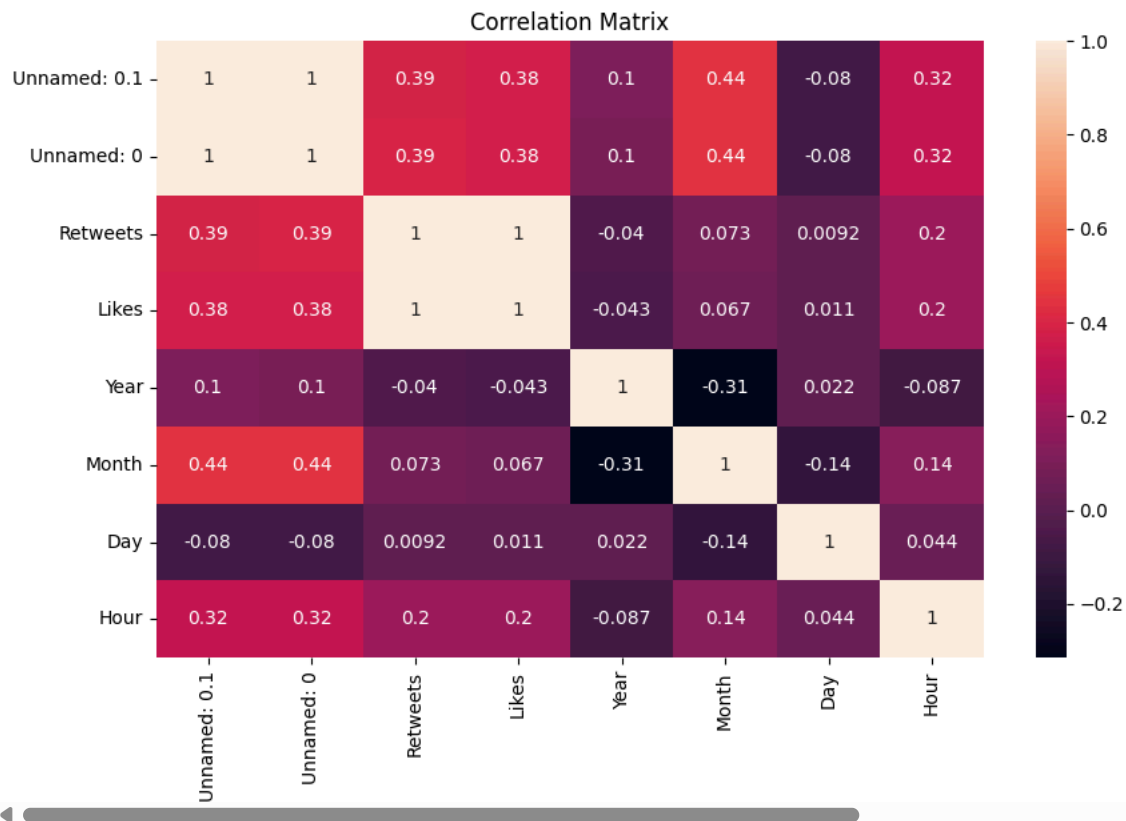
Year      -0.314845  0.021973 -0.087470
Month     1.000000 -0.135873  0.137835
Day       -0.135873  1.000000  0.044072
Hour      0.137835  0.044072  1.000000

```

```

plt.figure(figsize=(10, 6))
sns.heatmap(corr_matrix, annot=True)
plt.title("Correlation Matrix")
plt.show()

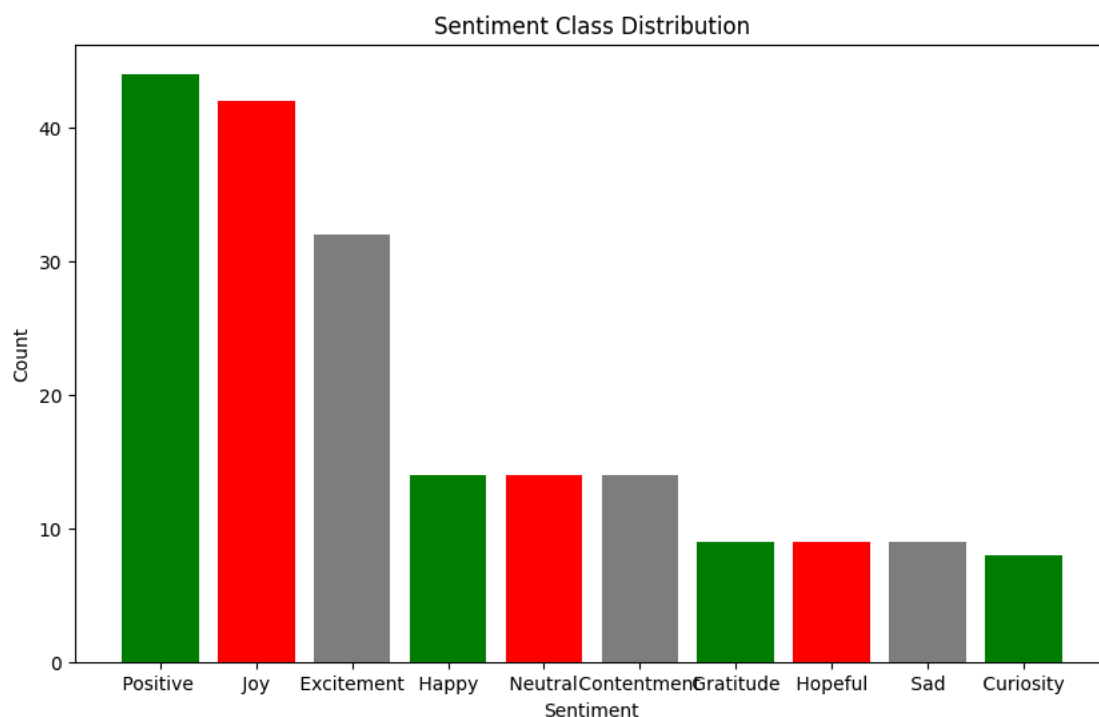
```



```

sentiment_ctr = df['Sentiment'].value_counts().head(10)
plt.figure(figsize=(10, 6))
plt.bar(sentiment_ctr.index, sentiment_ctr.values, color=['green', 'red', 'gray'])
plt.title('Sentiment Class Distribution')
plt.xlabel('Sentiment')
plt.ylabel('Count')
plt.show()

```

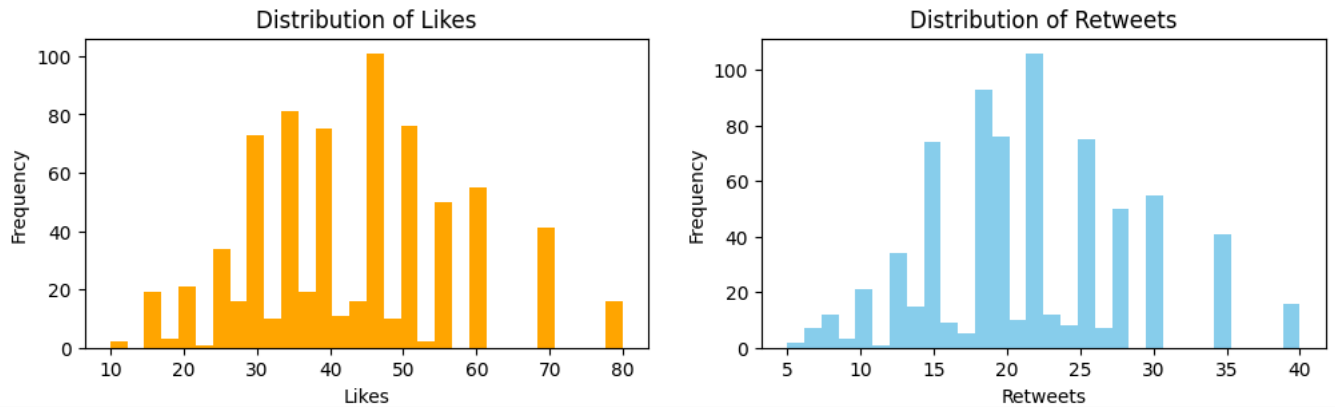


```
plt.figure(figsize=(12, 3))

plt.subplot(1, 2, 1)
plt.hist(df['Likes'].dropna(), color=['orange'], bins=30)
plt.title('Distribution of Likes')
plt.xlabel('Likes')
plt.ylabel('Frequency')

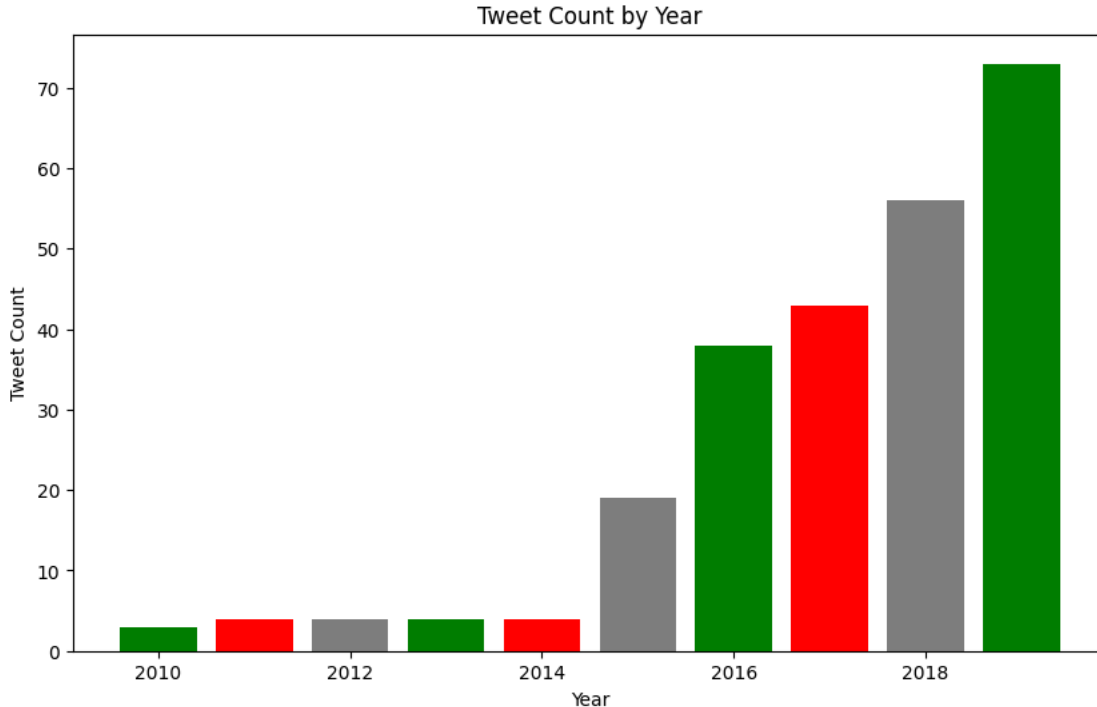
plt.subplot(1, 2, 2)
plt.hist(df['Retweets'].dropna(), color=['skyblue'], bins=30)
plt.title('Distribution of Retweets')
plt.xlabel('Retweets')
plt.ylabel('Frequency')
```

↻ Text(0, 0.5, 'Frequency')



```
year_ctr = df['Year'].value_counts().sort_index().head(10)
plt.figure(figsize=(10, 6))
plt.bar(year_ctr.index, year_ctr.values, color=['green', 'red', 'gray'])
plt.title('Tweet Count by Year')
plt.xlabel('Year')
plt.ylabel('Tweet Count')
plt.show()
```

↻



```
all_text = ' '.join(df['Text'].dropna().tolist())
wordcloud = WordCloud(width=1000, height=500, background_color='white').generate(all_text)

plt.figure(figsize=(15, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("Word Cloud of Tweet Texts")
plt.show()
```

